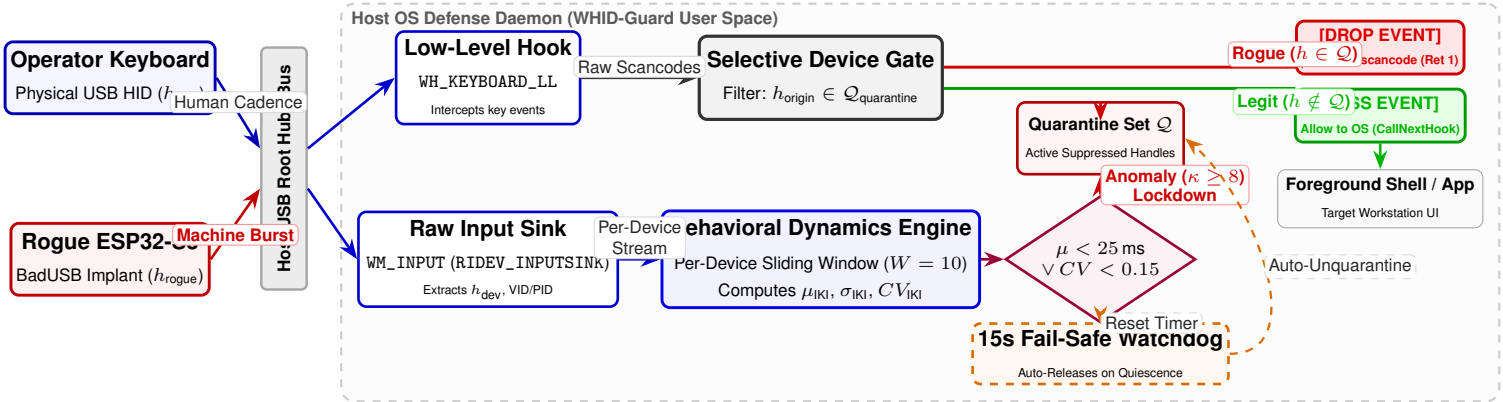

Host OS Defense Daemon (WHID-Guard User Space)
Operator Keyboard
Physical USB HID (h_{human})
Human Cadence
Rogue ESP32-S3
BadUSB Implant (h_{rogue})
Machine Burst
Host USB Root Hub
Low-Level Hook
WH_KEYBOARD_LL
Intercepts key events
Raw Scancodes
Selective Device Gate
Filter: $h_{\text{origin}} \in \mathcal{Q}_{\text{quarantine}}$
[DROP EVENT]
Rogue ($h \in \mathcal{Q}$) Scancode (Ret 1)
Legit ($h \notin \mathcal{Q}$) PASS EVENT]
Allow to OS (CallNextHook)
Foreground Shell / App
Target Workstation UI
Quarantine Set $\mathcal{Q}$
Active Suppressed Handles
Anomaly ($\kappa \geq 8$)
Lockdown
Raw Input Sink
WM_INPUT (RIDEV_INPUTSINK)
Extracts h_{dev}, VID/PID
Per-Device Stream
Behavioral Dynamics Engine
Per-Device Sliding Window ($W = 10$)
Computes μ_{IKI}, σ_{IKI}, CV_{IKI}
$\mu < 25\,\text{ms}$
$\vee\, CV < 0.15$
Auto-Unquarantine
Reset Timer
15s Fail-Safe Watchdog
Auto-Releases on Quiescence